

CS23532-COMPUTER NETWORKS-LAB MANUAL

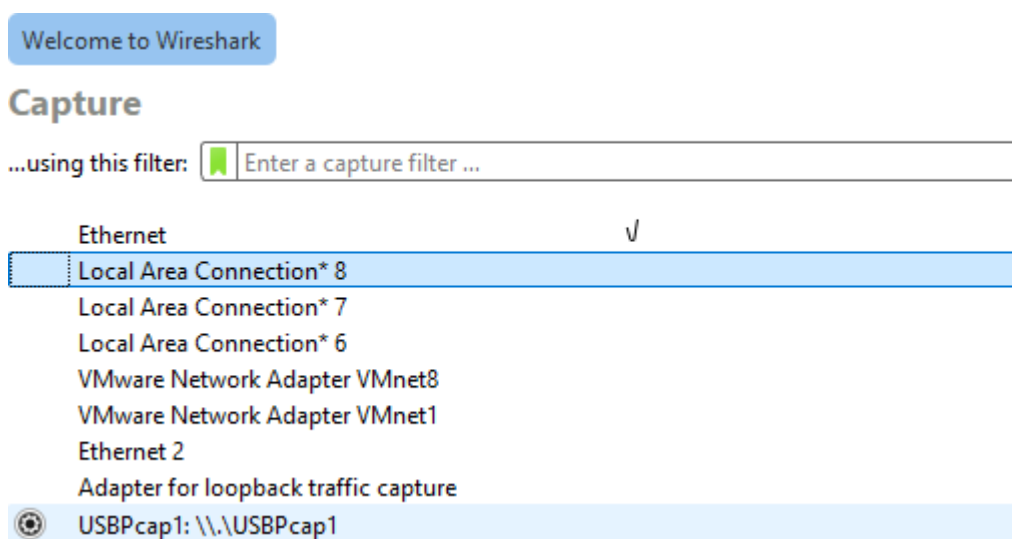
Practical-5

AIM Experiments on Packet capture tool: Wireshark

PROCEDURE

1. Download and install the **Wireshark packet capture tool** on the computer.
2. Open Wireshark and observe the available network interfaces.
3. Select the required active network interface, such as **Wi-Fi or Ethernet**.
4. Start the packet capture process to monitor the network traffic.
5. Generate network activity by browsing a website or performing network communication.
6. Stop the packet capture and observe the captured packets and their details.
7. Apply suitable filters to analyze different network protocols and study the captured network traffic.

SCREENSHOTS



No.	Time	Source	Destination	Protocol	Length	Info
455	4.029655	172.16.75.93	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
456	4.029786	172.16.75.93	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
457	4.034279	172.16.79.251	239.0.0.0	UDP	1078	39732 → 51721 Len=1036
458	4.039148	172.16.76.60	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::d98b:6c0e:45e6:adac A 172.16.76.60
459	4.039433	172.16.79.249	239.0.0.0	UDP	1078	53696 → 51721 Len=1036
460	4.039433	fe80::d98b:6c0e:45e6:adac	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::d98b:6c0e:45e6:adac A 172.16.76.60
461	4.065317	172.16.79.248	239.0.0.0	UDP	1078	60790 → 51721 Len=1036
462	4.065976	CloudNetwork_79:1f::	Broadcast	ARP	60	Who has 172.16.76.3? Tell 172.16.75.75
463	4.075041	172.16.79.251	239.0.0.0	UDP	1078	47989 → 51721 Len=1036
464	4.078653	fe80::cc8d:ce94:740c:dc37	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::cc8d:ce94:740c:dc37 A 172.16.76.144
465	4.078653	172.16.76.144	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::cc8d:ce94:740c:dc37 A 172.16.76.144
466	4.078984	ea:55:ac:41:ed:03	Broadcast	ARP	60	Who has 172.16.76.6? Tell 172.16.76.144
467	4.079358	fe80::cc8d:ce94:740c:dc37	ff02::1:ff4b:dc37	ICMPv6	86	Neighbor Solicitation for fe80::305e:847a:194b:dc37 from ea:55:ac:41:ed:03
468	4.081102	172.16.79.249	239.0.0.0	UDP	1078	40958 → 51721 Len=1036
469	4.093706	fe80::490b:1406:77db:69aa	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::490b:1406:77db:69aa A 172.16.76.90
470	4.093706	172.16.76.90	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::490b:1406:77db:69aa A 172.16.76.90
471	4.093706	fe80::490b:1406:77db:69aa	ff02::1:ff4b:dc37	ICMPv6	86	Neighbor Solicitation for fe80::305e:847a:194b:dc37 from ea:55:ac:41:ed:03
472	4.093706	CloudNetwork_78:8c::	Broadcast	ARP	60	Who has 172.16.76.6? Tell 172.16.76.90
473	4.094340	CloudNetwork_79:20::	Broadcast	LLC	60	S P, Func=RRR, N(R)=64; DSAP NULL LSAP Individual, SSAP NULL LSAP Command
474	4.094565	fe80::75b7:683d:23d3:c11b	ff02::fb	MDNS	139	Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
475	4.094565	172.16.76.112	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
476	4.094701	172.16.76.112	224.0.0.251	MDNS	262	Standard query response 0x0000 A 172.16.76.112 AAAA fe80::2b:ec15:d25e:468a TXT
477	4.094701	172.16.76.112	224.0.0.251	MDNS	262	Standard query response 0x0000 A 172.16.76.112 AAAA fe80::2b:ec15:d25e:468a TXT
478	4.094810	172.16.76.112	224.0.0.251	MDNS	119	Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112

No.	Time	Source	Destination	Protocol	Length	Info
1	0.869006	ClooudNetwork_78:01::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.49
2	0.842647	ClooudNetwork_77:ff::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.55
3	0.909504	ClooudNetwork_78:01::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.49
4	2.113611	ClooudNetwork_77:ff::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.55
5	1.697712	MNC_73:05:00	Broadcast	ARP	60	ARP Announcement for 172.16.79.143
6	1.977227	ClooudNetwork_78:01::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.49
7	2.113623	ClooudNetwork_78:79::	Broadcast	ARP	60	ARP Announcement for 0.0.0.0
8	2.122148	XiaomiCom_S4:0D::	Broadcast	ARP	60	ARP Announcement for 0.0.0.0
9	2.158382	x2f82af71:12fe	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.79.139
10	2.513752	ClooudNetwork_77:ff::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.55
11	2.406765	ClooudNetwork_78:70::	Broadcast	ARP	60	Who has 172.16.76.182? (ARP Probe)
12	2.471897	ClooudNetwork_77:ff::	Broadcast	ARP	60	Who has 172.16.76.182? Tell 172.16.76.55
13	2.502144	7a5c08b133:4d6e	Broadcast	ARP	60	Who has 172.16.76.182? Tell 172.16.76.92
14	2.582246	ClooudNetwork_78:70::	Broadcast	ARP	60	Who has 172.16.76.182? Tell 172.16.76.97
15	2.595513	42:b739:0a:03:09	Broadcast	ARP	60	Who has 172.16.76.182? Tell 172.16.76.54
16	2.735569	be1a41ca:2d58	Broadcast	ARP	60	Who has 172.16.76.182? Tell 172.16.76.53
17	2.744638	XiaomiCom_S4:0D::	Broadcast	ARP	60	ARP Announcement for 0.0.0.0
18	2.977836	ClooudNetwork_78:70::	Broadcast	ARP	60	ARP Announcement for 0.0.0.0
19	2.984878	ClooudNetwork_78:01::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.49
20	3.322185	ClooudNetwork_77:ff::	Broadcast	ARP	60	Who has 172.16.76.182? Tell 172.16.76.55
21	3.423633	ClooudNetwork_78:79::	Broadcast	ARP	60	Who has 172.16.76.182? (ARP Probe)
22	3.512738	ClooudNetwork_77:ff::	Broadcast	ARP	60	Who has 172.16.72.1? Tell 172.16.76.55
23	3.583626	Sophos_ff:ba1e	Broadcast	ARP	60	Who has 172.16.76.183? Tell 172.16.72.1
24	3.695660	82:40:2a:5d:1b:2a	Broadcast	ARP	60	ARP Announcement for 0.0.0.0

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface Device\NPF_{5A53B0E8-3E45-4EC6-B00F-4E980895}

Ethernet II, Src: ClooudNetwork_78:01:41:43:02:82(a9:78:01:43), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (Request)

```

0000 ff ff ff ff ff ff ac 82 a9 78 01 43 02 80 00 01 .....x.C...
0001 00 00 00 00 00 00 ac 82 a9 78 01 43 02 80 45 .....x.C..E
0002 00 00 00 00 00 00 ac 10 43 01 00 00 00 00 00 .....H.....
    
```

Result

Thus, the **Wireshark** packet capture tool was successfully used to capture, filter, and analyze network packets transmitted through the selected network interface.